Heaps — Priority Que

- Minimizing Heap
- Maximizing Heap

NB: Has No pointers

Heap Sort

if n elements

0 Sort(list) => 0 = 1

1 2

3 4 5 6

7 8 9 10 11 12 13 14

n...

How deep is the tree.

$\log_2(n)$

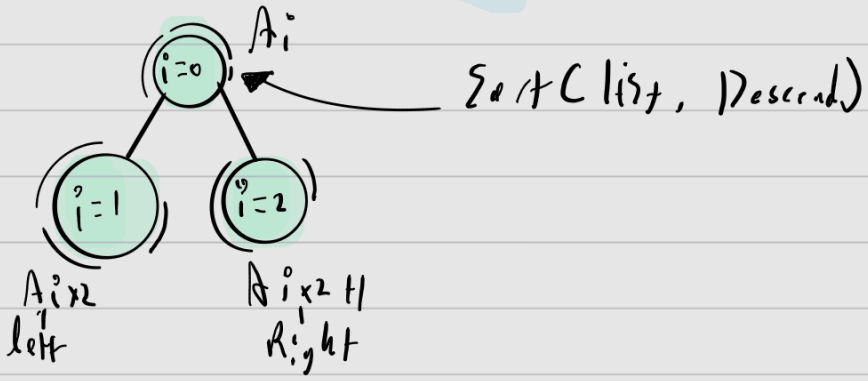
- Prove depth is $\log_2(n)$
- Quick sort: Tony Hall → in Moscow. Trying to sort russian dictionary.
- John Backus: 2.5 years to make Fortran Compiler.
- Compilers have to sort to run.

Characteristics of Heaps

- "Most import element" at top
- Store heap in Array Node[]
- i) LeftChild(i) = 2 x i
  ii) RightChild(i) = 2 x i + 1
  iii) Parent(i) = i ÷ 2

Maximising Heap:

Greatest element on Top



$\Rightarrow A_i > A_{i \times 2} \;\&\&\; A_i > A_{i \times 2 + 1}$

Heap Methods

- Node removeMax(); or removeMin();
- boolean isEmpty();
- boolean isFull();
- void insert(Node n);
- buildUp(Node[], int sizeX)

Heapify